

PROOF OF CONCEPT

Snowflake Data Lineage

Internal & external lineage — capabilities, boundaries and PoC findings

July 2026

01

Scope & Approach

02

PoC Architecture

03

Capabilities & Limitations

04

Findings & Conclusion

tieto

01 PoC Scope & Approach

OBJECTIVE Evaluate Snowflake's lineage capability — in-platform and cross-platform (Airflow, ADF, OpenLineage supported tools) — at table and column granularity across a Bronze → Silver → Gold medallion.

IN SCOPE

- Snowflake native lineage via Snowsight + GET_LINEAGE()
- Table & column level across the 3-layers in medallion
- ADF ingestion: XML file + Azure SQL; JSON file
- COPY INTO from internal stage; stored-proc transforms
- External orchestrator(Airflow) lineage
- Record level lineage in Snowflake using CoCo
- Document Snowflake lineage capabilities & limitations

OUT OF SCOPE

- Integrity dataset / production data
- Real stored-procedure logic (dummy procs used)
- Enterprise rollout & production deployment
- Performance & credit-cost optimization

WHAT WE DID

- Provisioned Bronze/Silver/Gold layers in Snowflake sandbox Enterprise edition
- ADF & Copy into based data ingestion
- Raw → Refined → Curated: CTAS, INSERT...SELECT, MERGE Stored procedures, Snowpark
- Permissions set up for lineage
- Leveraged CoCo for record level lineage
- Validated table + column lineage & proc attribution
- Probed external lineage; logged gaps

4-WEEKS EXECUTION

1

WEEK 1

Discover & Decide

Feature research (native + external),
license / sandbox check, scope

2

WEEK 1

Build the Pipeline

ADF ingestion, internal stage & 3-layer
medallion model

3

WEEK 2-3

Capture & Validate

Enable lineage; validate table + column,
native + external

4

WEEK 4

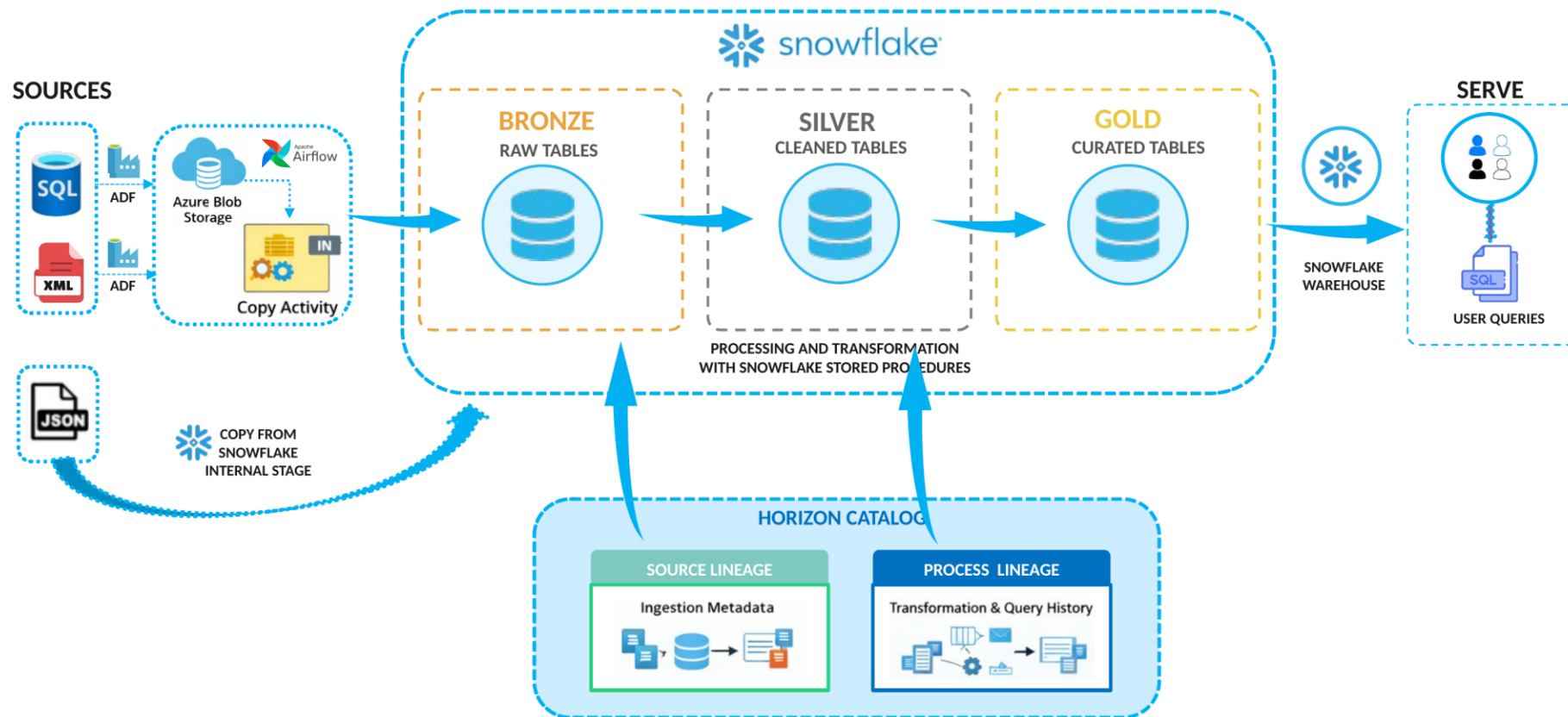
Wrap-Up

Document findings, limitations & demo

02

PoC Architecture

ADF lands files + an Azure SQL source into an external stage; **COPY INTO** loads Bronze, and stored procedures promote data through Silver to Gold — with **Horizon Catalog** capturing source & process lineage.



CAPABILITIES

- **Native lineage on Enterprise Edition**

Snowsight → Catalog → Database Explorer → Lineage tab; programmatic via `SNOWFLAKE.CORE.GET_LINEAGE()`. Table & column level.

- **Broad object coverage**

Tables, Iceberg, views, materialized & semantic views, stages, ML datasets / feature views / models.

- **Operations & dependencies**

Data movement (`COPY INTO`, `CTAS`, `CLONE`, `CREATE VIEW/MV`, `INSERT...SELECT`, `MERGE`, cross-table `UPDATE`) + object dependencies.

- **External & ML lineage**

OpenLineage push-model for external ETL & databases; ML lineage.

- **Horizon Catalog (Summit 2026)**

Select Star metadata connectors + automated cross-platform, column-level lineage via OpenLineage API — Private Preview.

- **Access & retention**

`VIEW LINEAGE` (PUBLIC default)

LIMITATIONS

- **Excluded objects**

No lineage for shared DBs or any `INFORMATION_SCHEMA`; temp tables never shown

- **Graph behaviour**

Dropped tables drop out (renamed remain)

- **Stored-procedure limits**

Disjointed multi-statement moves (`SET var = (SELECT...)` then `INSERT...VALUES`) break lineage; `GET_LINEAGE()` can't return procedure lineage — UI only.

- **Column & procedure detail**

Procedure name & code surfaced, but not individual columns; external lineage has no column-level lineage.

- **External constraints**

One side must be a Snowflake object; OpenLineage v2 unsupported.

- **Coverage cut-offs**

Semantic views created before Feb 2026 excluded.

INTERNAL LINEAGE

Table-level lineage

Snowsight + GET_LINEAGE()

Supported

Column-level lineage

Native, in-platform objects

Supported

Internal stage → table (COPY INTO)

Native, in-platform, table level with copy command

Partial

Procedure logic

Proc name & code; not per-column; UI only

Partial

Record level lineage

Using CoCo, shows lineage within Snowflake objects

Partial

EXTERNAL LINEAGE

Snowflake ↔ dbt

OpenLineage; documented

Supported

Snowflake ↔ Astronomer Airflow

Tested, Shows the code but not Airflow in the lineage

Partial

Snowflake ↔ OpenFlow (Azure SQL)

Tested, generates table level lineage

Partial

External stage (Blob)

Tested, generates table level lineage

Partial

Snowflake ↔ ADF

Tested, Limited OpenLineage support

Not supported

Snowflake ↔ Nexla

Not tested

Unknown

CONCLUSION

Native internal lineage (table + column) is production-ready and met table & column lineage but without the individual column transformation. **External / cross-platform lineage is still maturing** — dbt is supported as it has native support, ADF and Nexla are unsupported today, and several paths remain to be tested. Once Select Star is fully integrated & GA, additional external lineage capabilities will get enhanced.

Recommendation: adopt native lineage for in-platform coverage, and pilot Horizon Catalog (Select Star) connectors (when available) to validate the external, column-level lineage generation.

Demo

Thank you

tieto